Deepanshu Lakde

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Summary

AI Backend Engineer with ~1.5 years of experience building asynchronous FastAPI services, data ingestion pipelines and multi-LLM orchestration (Claude, OpenAI, Gemini) with LangGraph. Experienced in taking LLM workflows to production: cost optimization, fallback handling, enterprise SSO integration and CI/CD-tested microservices.

Experience

Jr. AI Backend Engineer Jan 2026 – Present

Anervea-AI *Pune, MH*

- Engineered a highly concurrent asynchronous ingestion pipeline using httpx and Playwright, processing 50,000+ clinical records daily with zero downtime.
- Reduced LLM token consumption by 30% while maintaining extraction quality by orchestrating LangGraph workflows with Redis-backed state persistence, context-aware routing, and fallback handling across Claude, OpenAI, and Gemini.
- Integrated OpenID Connect (OIDC) Single Sign-On for enterprise clients, enabling secure single sign-on into the platform.
- Achieved 85% test coverage on asynchronous FastAPI microservices, ensuring deployment reliability by integrating Pytest within a Jenkins CI/CD pipeline.

Backend Python Intern Jan 2025 – Jul 2025

Assimilate Technologies Pvt. Ltd. *Pune, MH*

- Reduced core API latency by 25% (1100ms → 820ms) by optimizing PostgreSQL queries and implementing strategic indexing within the Horilla HRMS backend.
- Supported 10,000+ concurrent user requests with no data-integrity incidents by developing FastAPI microservices with automated validation and strict idempotency handling.

Projects

Agentic SQL Analytics Gateway | *LangGraph, FastAPI, PostgreSQL, pgvector, Docker, Redis*

- Architected a multi-agent LangGraph gateway exposing async FastAPI endpoints to translate natural language queries into optimized, execution-safe PostgreSQL and pgvector plans.
- Implemented context-aware routing, automated schema retrieval, and LLM fallback strategies (Claude/OpenAI), reducing complex relational query failure rates by 35%.
- Containerized the gateway with Docker Compose, managing multi-turn conversational state persistence via Redis for self-serve analytics.

ERPNext AI Resume Screening Plugin | *Python, Frappe, MariaDB, Sentence Transformers*

- Accelerated candidate shortlisting time by 75% by deploying a custom AI screening plugin for an enterprise ERPNext system, integrating semantic matching directly into existing HR workflows.
- Developed secure backend REST APIs via the Frappe framework, extending MariaDB schemas to manage dense vector embeddings while enforcing strict role-based access controls.
- Engineered PDF parsing heuristics and integrated Sentence Transformers, optimizing cosine similarity logic to deliver sub-second candidate recommendations.

Alfakinetic Competitive Intelligence Platform | *FastAPI, Playwright, PostgreSQL, MongoDB*

- Reduced extraction latency by 40% by engineering an asynchronous FastAPI backend to orchestrate Playwright scraping alongside structured LLM extraction workflows.
- Maintained sub-50ms retrieval latency across 15 million records with a hybrid database topology: PostgreSQL for relational transactions and MongoDB for high-volume unstructured signals.

Technical Skills

Languages: Python, SQL, JavaScript

AI & LLM: OpenAI, Claude, Gemini, LangGraph, Agentic AI, RAG, Embeddings, Voyage AI, Prompt Engineering

Backend: FastAPI, Flask, Django, REST APIs, AsyncIO

ML & NLP: PyTorch, TensorFlow, scikit-learn, Sentence Transformers, Hugging Face

Databases: PostgreSQL, MongoDB, Redis, MySQL, pgvector, Pinecone

Infrastructure: Docker, Docker Compose, Jenkins CI/CD, AWS, Git, uv, Pytest

Education

Master of Computer Applications (MCA) | 7.99 CGPA Aug 2023 – Jul 2025

PR Pote Patil Institute of Engineering and Management *Amravati, MH*

Bachelor of Computer Application (BCA) | 78.75% Aug 2020 – Jul 2023

Sant Gadge Baba Amravati University *Amravati, MH*